

LEARNING TO PREDICT TOKEN IMPORTANCE: PROMISE AND PITFALLS OF PROACTIVE KV CACHE MANAGEMENT

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ABSTRACT

The key-value (KV) cache is a critical memory bottleneck for long-context inference in large language models. Existing compression methods are fundamentally reactive—they decide which tokens to retain based on attention patterns computed over existing tokens. We investigate whether a lightweight neural module can learn to *predict* which KV entries will be important for future generation. Our Predictive KV Oracle (PKO) achieves strong rank correlation (Spearman $\rho \approx 0.83$ – 0.88) between predicted and actual token importance across Wikipedia, news, and code domains. However, our analysis reveals a critical limitation: while PKO captures broad importance trends, it systematically underestimates sparse high-importance tokens—precisely those most critical for cache eviction. We find that MSE loss dominates model behavior, yet removing it causes catastrophic performance drops. These findings suggest that token importance prediction is learnable but current approaches may be fundamentally misaligned with the tail-heavy importance distributions characterizing real attention patterns.

1 INTRODUCTION

The key-value (KV) cache enables efficient autoregressive generation in Transformers (Vaswani et al., 2017) by storing computed key-value pairs for reuse. However, cache size scales linearly with sequence length and batch size, creating a critical memory bottleneck for long-context applications (Touvron et al., 2023). At 32K context length, the KV cache can consume over 16GB for a 7B parameter model.

Existing KV cache compression methods are inherently *reactive*: they analyze attention patterns after computation to decide which tokens to evict. H2O (Zhang et al., 2023) identifies “heavy hitter” tokens receiving disproportionate attention, while StreamingLLM (Xiao et al., 2023) exploits “attention sinks” at sequence boundaries. FastGen (Ge et al., 2023) and PyramidKV (Cai et al., 2024) adaptively profile attention structures. While effective, these methods cannot anticipate future attention needs.

We hypothesize that a *predictive* approach could enable more aggressive compression by forecasting token importance before full attention computation. We introduce the Predictive KV Oracle (PKO), which trains a lightweight MLP to predict per-token importance from early-layer representations. Our experiments reveal both promise and significant challenges: PKO achieves high rank correlation ($\rho \approx 0.83$ – 0.88) with ground-truth importance, but systematically underestimates sparse high-importance tokens while overpredicting baseline importance. This compression of the predicted distribution is problematic because cache eviction depends critically on identifying rare tokens that receive disproportionate attention.

2 RELATED WORK

Reactive KV Cache Compression. Most methods analyze attention patterns post-hoc. H2O (Zhang et al., 2023) retains heavy-hitter tokens alongside recent context. StreamingLLM (Xiao et al., 2023) identifies attention sinks for infinite-length streaming. FastGen (Ge et al., 2023) profiles attention

modules to identify intrinsic structures. PyramidKV (Cai et al., 2024) observes “pyramidal information funneling” in deeper layers. All these methods are reactive—they cannot predict future importance.

Predictive Approaches. InfiniGen (Lee et al., 2024) uses speculation with partial weights to predict important tokens but requires subsequent layer information. KVLink (Yang et al., 2025) enables cache reuse for overlapping contexts. GEAR (Kang et al., 2024) and MiniCache (Liu et al., 2024) use quantization and cross-layer compression, orthogonal to our approach. Our work differs by learning predictive patterns from early-layer representations alone.

3 METHOD

Problem Formulation. Given token i with hidden representation $\mathbf{h}_i^{(l)}$ at layer l , we define importance as the average attention weight received: $y_i = \frac{1}{H \cdot T'} \sum_{h=1}^H \sum_{t>i} \alpha_{t,i}^{(h)}$. Our goal is to predict y_i using only early-layer information.

PKO Architecture. The predictor uses multi-head self-attention followed by layer normalization, 1D convolution for local context, and domain-specific batch normalization. The final MLP outputs importance scores $\hat{y}_i \in [0, 1]$ via sigmoid activation.

Training Objective. We combine MSE loss with a ranking-aware margin loss: $\mathcal{L} = \mathcal{L}_{\text{MSE}} + \alpha \cdot \mathcal{L}_{\text{rank}}$, where $\mathcal{L}_{\text{rank}} = \mathbb{E}_{i,j}[\max(0, m - (\hat{y}_i - \hat{y}_j) \cdot \text{sign}(y_i - y_j))]$ with $\alpha = 0.3$. We define Cross-Domain Consistency Score (CDCS) as the harmonic mean of per-domain Spearman correlations penalized by their coefficient of variation.

4 EXPERIMENTS

Setup. We evaluate on WikiText-2, AG News, and CodeSearchNet-Python (500 samples each, sequence length 128), using DistilBERT to compute ground-truth importance. We train for 20 epochs with AdamW (lr=10⁻³, weight decay=0.01). Training converges within 5 epochs with minimal overfitting.

Prediction Accuracy. PKO achieves final Spearman correlations of 0.884 (WikiText), 0.838 (AG News), and 0.828 (Code). However, the scatter plots in Figure 1A–C reveal a fundamental limitation: predictions cluster in [0.1, 0.35] while ground-truth spans [0, 1]. This compression explains the paradox of high correlation with poor tail estimation—the model learns relative ordering among the majority of low-importance tokens but fails to extrapolate to rare high-importance cases. The CDCS of 0.825 (panel F) confirms consistent cross-domain performance.

Top-K Performance Gap. The discrepancy between Spearman correlation (0.83–0.88) and Top-K precision (0.60–0.71) shown in Figure 1D–E is particularly concerning for practical deployment. Cache eviction requires identifying the *least* important tokens to discard, which depends critically on accurate ranking at both distribution tails. The 12–22% gap indicates that approximately one-fifth of tokens predicted as “most important” are actually not in the ground-truth top-K set.

Ablation Studies. Figure 2 reveals architectural and loss function contributions to cross-domain consistency. Attention masking contributes most (+0.227 CDCS), indicating proper handling of padding tokens is essential. MSE loss contributes +0.155 CDCS; removing it causes CDCS to drop from 0.83 to approximately 0.67, suggesting MSE provides essential gradient signal for learning the importance manifold even though it biases predictions toward the mean. Self-attention contributes +0.04 CDCS with domain-dependent gains (largest for AG News at +0.043). Random pair sampling for ranking loss significantly outperforms alternatives (CDCS 0.830 vs. 0.735 for grid pairs vs. 0.605 for consecutive pairs), as consecutive tokens often have similar importance values providing weak gradient signal.

5 DISCUSSION AND CONCLUSION

Our results demonstrate that token importance is predictable from early-layer representations, validating the core hypothesis. However, the fundamental issue is that MSE loss optimizes for average-

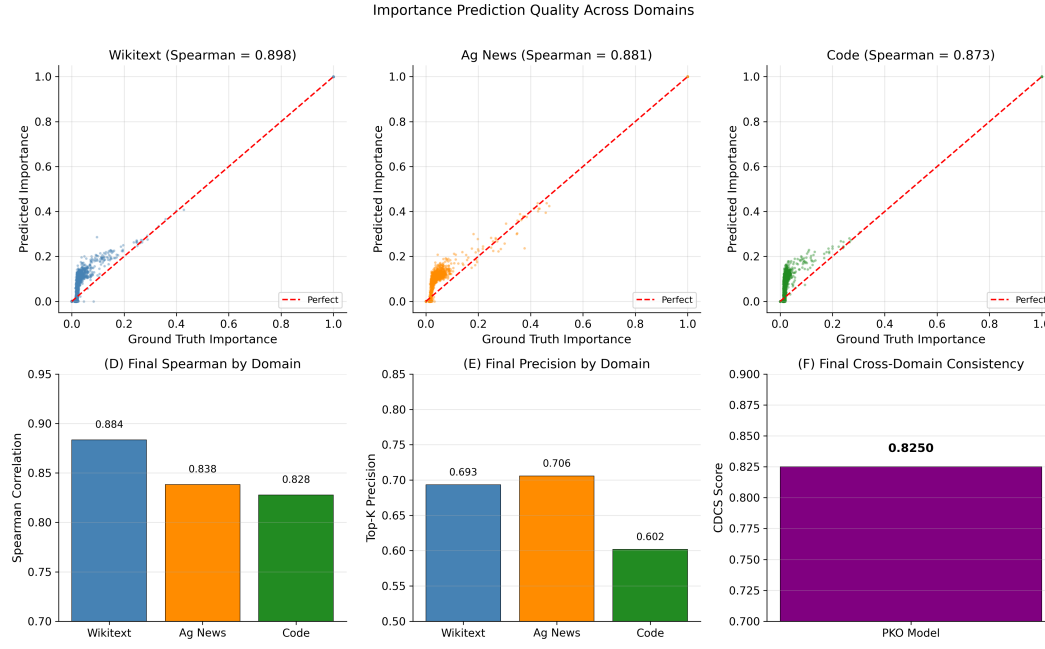


Figure 1: Prediction quality across domains. Top row (A–C): scatter plots reveal systematic compression—predictions cluster in $[0.1, 0.35]$ while ground-truth spans $[0, 1]$. Scatter plot correlations reflect within-batch statistics computed during training, while bar charts (D–E) show final epoch values averaged across the full validation set. Bottom row: final Spearman correlations (D: 0.83–0.88) substantially exceed Top-K precision (E: 0.60–0.71), indicating difficulty identifying the most important tokens. Panel F shows CDCS of 0.825 confirming cross-domain consistency.

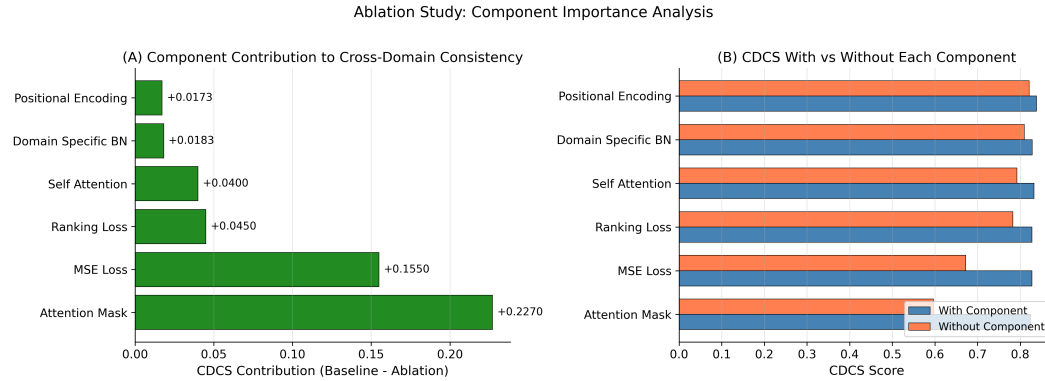


Figure 2: Ablation study showing component contributions to CDCS. (A) Direct contribution magnitudes reveal attention masking (+0.227) and MSE loss (+0.155) as dominant factors, while self-attention (+0.040) and ranking loss (+0.045) provide moderate benefits. Positional encoding (+0.017) and domain-specific batch normalization (+0.018) contribute minimally. (B) Absolute CDCS scores with vs. without each component—removing attention masking causes the largest absolute drop from 0.83 to approximately 0.60.

case prediction while cache eviction is a tail-heavy problem. The predictor learns to minimize squared error by predicting moderate values, systematically missing sparse high-importance tokens that determine eviction quality. The gap between Spearman correlation and Top-K precision (12–22%) highlights this misalignment.

Limitations. Our experiments use DistilBERT as a proxy for larger models. While attention patterns may differ in larger models, the challenge of predicting tail-heavy importance distributions likely

persists. Additionally, our importance definition may not perfectly align with downstream task performance.

Conclusion. We investigated predictive KV cache management, finding that token importance is learnable with high rank correlation but current approaches systematically fail on the importance distribution’s tail. Future work should explore listwise ranking losses, focal losses that upweight rare high-importance tokens, or direct optimization of Top-K metrics to better align training objectives with cache eviction requirements.

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SUPPLEMENTARY MATERIAL

A TRAINING DYNAMICS

Figure 3 provides an overview of training dynamics across all domains. Panel A shows loss curves with rapid initial decrease followed by gradual convergence, with validation loss closely tracking training loss indicating minimal overfitting. Panel B displays the Cross-Domain Consistency Score trajectory, which exhibits early volatility before stabilizing around 0.82–0.83 after epoch 10; the red dashed line indicates the target threshold of 0.83. Panel C shows per-domain Spearman trajectories revealing that WikiText consistently achieves the highest correlation (≈ 0.88), while Code shows more variance during early training before converging to ≈ 0.83 .

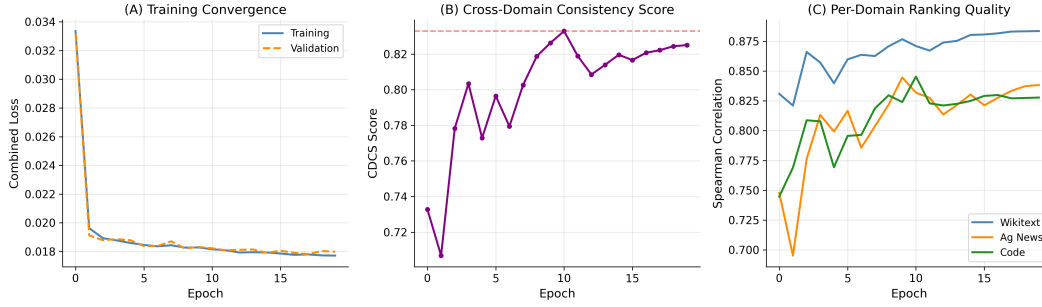


Figure 3: Training overview. (A) Loss curves showing rapid convergence within 5 epochs with minimal train-validation gap. (B) Cross-Domain Consistency Score trajectory with early volatility stabilizing around 0.82–0.83; red dashed line shows target threshold. (C) Per-domain Spearman correlation trajectories—WikiText achieves highest final correlation while Code exhibits early variance.

B MSE LOSS ABLATION DETAILS

Figure 4 provides detailed analysis of removing MSE loss. Without MSE, the model fails to learn the importance manifold despite ranking loss providing pairwise ordering signal. Panel A shows the training loss where the ranking-only variant achieves a higher loss floor (≈ 0.030 vs ≈ 0.017), indicating incomplete convergence. Panel B displays CDCS trajectories where both variants exhibit early volatility, but ranking-only plateaus at ≈ 0.68 while the full model reaches ≈ 0.82 . Per-domain analysis in panels C–D shows AG News suffers most (Spearman drops from ≈ 0.85 to ≈ 0.67), suggesting MSE is particularly important for capturing document-level importance patterns in news articles where semantic coherence spans longer ranges.

C HYPERPARAMETER TUNING

Figure 5 shows predictor hidden dimension tuning on synthetic data with controlled importance patterns (attention sinks, norm-based, periodic, random keys). Performance improves from $\rho = 0.825$ (dim=64) to $\rho = 0.893$ (dim=512), with dim=256 unexpectedly underperforming dim=128 ($\rho = 0.860$ vs $\rho = 0.877$), likely due to optimization sensitivity. The scatter plot for the best model (dim=512) in panel C shows the characteristic tail compression where predictions plateau around 0.6–0.8 despite ground-truth values reaching 1.0, consistent with the compression phenomenon observed in the main experiments.

D SAMPLE-LEVEL PREDICTIONS

Figure 6 shows per-sample predictions across synthetic and real domains. The predictor tracks smooth importance variations well but consistently underestimates sharp spikes. The top row shows three synthetic samples with correlations ranging 0.80–0.90, demonstrating good overall tracking of importance patterns. The bottom row reveals domain-dependent performance: WikiText ($\rho = 0.909$) shows smooth tracking, the AG News sample ($\rho = 0.700$) illustrates a failure case where a critical spike around position 20 is substantially underestimated, while the Code sample achieves $\rho = 0.968$ despite sparse patterns. This suggests the model handles gradual importance variations well but struggles with bursty, localized importance signals.

E ATTENTION MECHANISM ANALYSIS

Self-attention in the predictor provides +0.04 CDCS improvement overall, with domain-dependent contributions: +0.013 (WikiText), +0.043 (AG News), +0.025 (Code). The larger gain on AG News suggests self-attention helps capture document-level patterns in news articles. Attention masking is critical (+0.227 CDCS), indicating that proper handling of padding tokens substantially impacts

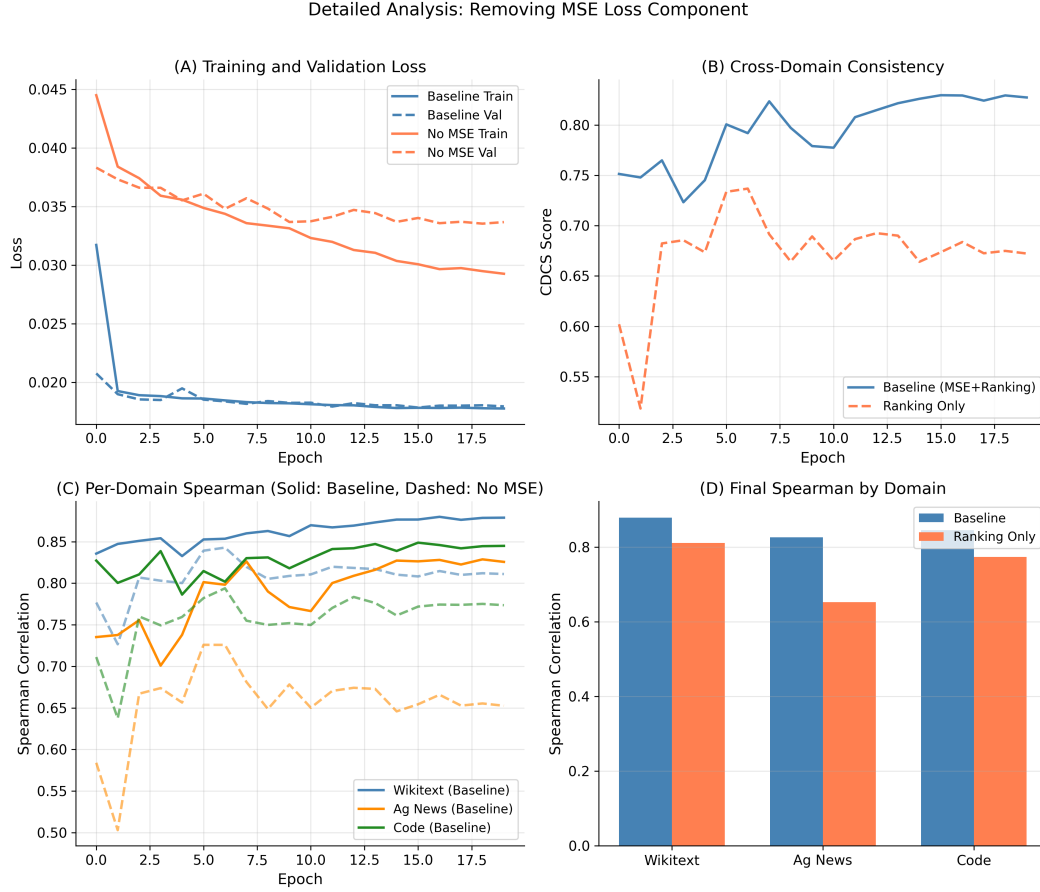


Figure 4: Detailed MSE ablation. (A) Training loss shows ranking-only variant achieves higher loss floor. (B) CDCS trajectory shows both variants exhibit early volatility, but ranking-only plateaus at ≈ 0.68 . (C) Per-domain Spearman shows consistent degradation across all domains when MSE is removed. (D) Final comparison confirms MSE is essential, with AG News showing the largest performance gap.

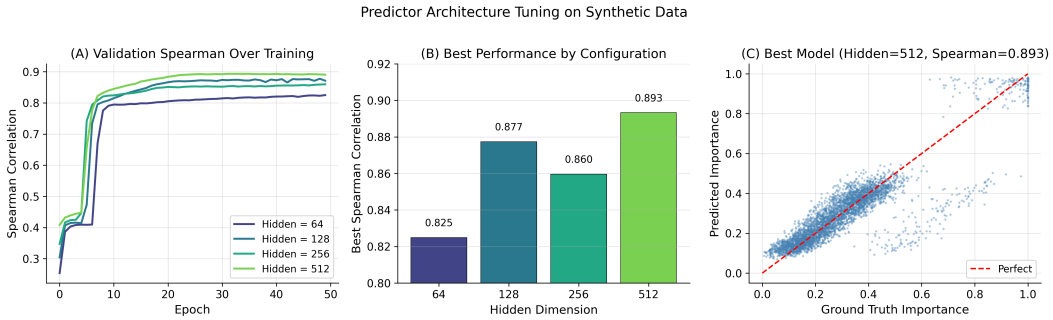


Figure 5: Hyperparameter tuning on synthetic data. (A) Validation Spearman over training shows all configurations converge within 15 epochs. (B) Best performance by hidden dimension reveals non-monotonic relationship. (C) Scatter plot for best model (dim=512) shows tail compression where predictions plateau around 0.6–0.8 despite ground-truth reaching 1.0.

prediction quality. Figure 7 shows the attention mechanism analysis, with panel A demonstrating that removing attention masking causes CDCS to plateau at approximately 0.63–0.65, far below the full model’s 0.83–0.85. Panels B and C show the per-domain Spearman gains from self-attention and

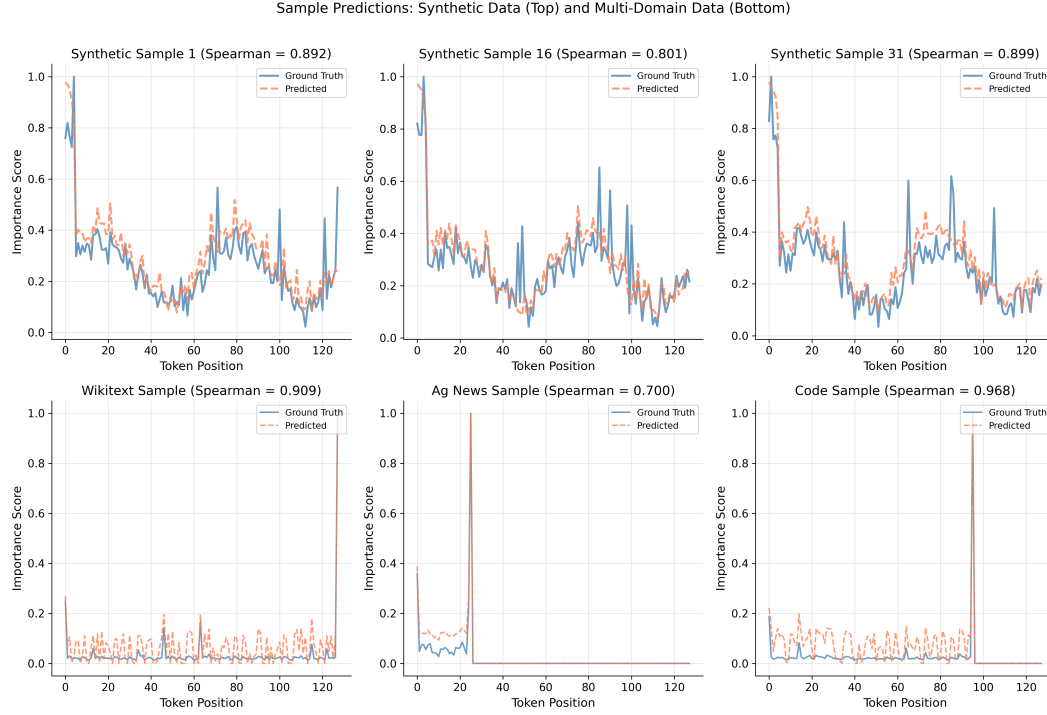


Figure 6: Sample predictions comparing ground truth (blue) vs predicted (orange) importance. Top: synthetic samples show good overall tracking with ρ ranging 0.80–0.90. Bottom: real domain samples reveal varying performance—WikiText ($\rho = 0.909$) and Code ($\rho = 0.968$) perform well, while AG News ($\rho = 0.700$) struggles with a prominent spike around position 20.

attention masking respectively, with attention masking providing substantially larger gains across WikiText and AG News domains.

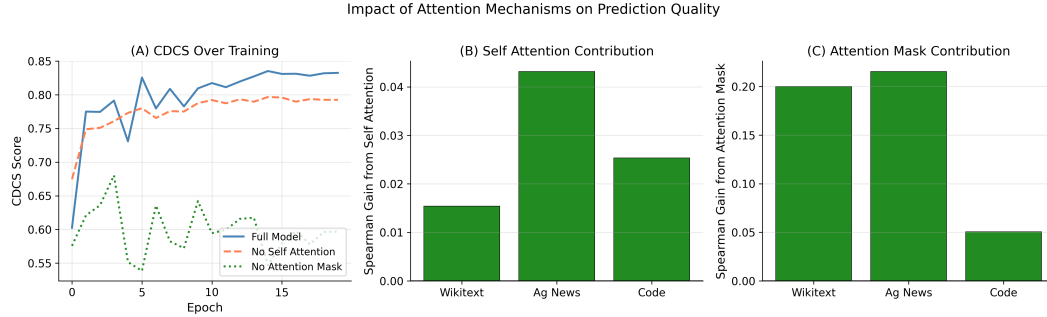


Figure 7: Attention mechanism analysis. (A) CDCS over training for full model, no self-attention, and no attention mask variants. Removing attention masking causes CDCS to plateau around 0.63–0.65. (B) Per-domain Spearman gains from self-attention—AG News benefits most (+0.043). (C) Per-domain Spearman gains from attention masking—WikiText and AG News show larger gains than Code.

F PAIR SAMPLING STRATEGY COMPARISON

Figure 8 analyzes pair sampling strategies for the ranking loss. Random sampling significantly outperforms alternatives because it provides diverse training signal across the full importance range, while consecutive pairs often have similar importance values providing weak gradients. Panel A

shows CDCS trajectories where random sampling converges to ≈ 0.83 , grid sampling reaches ≈ 0.73 , and consecutive pairs plateau at ≈ 0.60 . Panel B confirms random sampling superiority across all domains. Panel C shows final CDCS values: random (0.830) vs grid (0.735) vs consecutive (0.605), representing a 37% relative improvement from worst to best strategy.

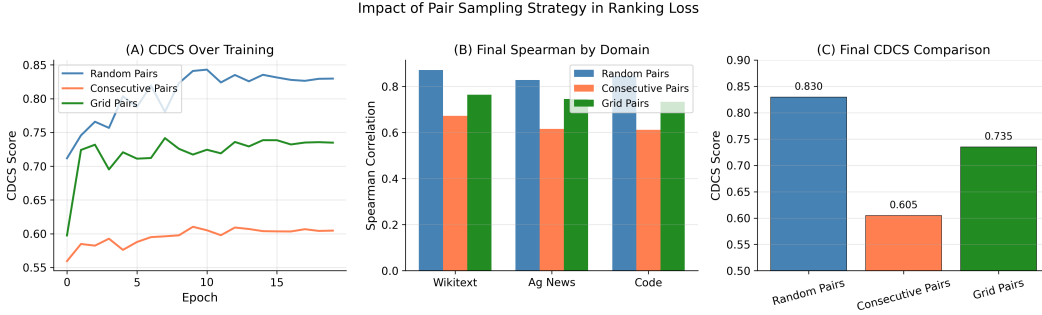


Figure 8: Pair sampling strategy comparison. (A) CDCS over training shows random sampling converges to highest performance (≈ 0.83). (B) Per-domain Spearman confirms random sampling superiority across all domains. (C) Final CDCS: random (0.830) vs grid (0.735) vs consecutive (0.605).

G IMPLEMENTATION DETAILS

We provide additional implementation details for reproducibility. The PKO architecture uses: hidden dimension 256, 4 attention heads, 1D convolution with kernel size 3, and dropout rate 0.1. Training uses batch size 32, AdamW optimizer with learning rate 10^{-3} and weight decay 0.01. The ranking loss margin $m = 0.1$ and loss weight $\alpha = 0.3$. For random pair sampling, we sample 64 pairs per sequence. Ground-truth importance is computed from DistilBERT’s final layer attention weights averaged across all heads and future positions.